

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0509

### COURSE OUTCOME COVERED

**CO4:** Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage:20%

**QUESTIONS: 5 Total Marks:50**

### Annexure A Industry Problem Solving Task

#### Code of Conduct:

I **N.AKSHAYA(192425129)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

#### Signature of the student:

| Q. No             | Understanding of Industry Problem (20%) | Application of Engineering Concepts (25%) | Solution Design (25%) | Use of Tools (15%) | Analysis (10%) | Professionalism (5%) | Total Score (100%) |
|-------------------|-----------------------------------------|-------------------------------------------|-----------------------|--------------------|----------------|----------------------|--------------------|
| 1                 |                                         |                                           |                       |                    |                |                      |                    |
| 2                 |                                         |                                           |                       |                    |                |                      |                    |
| 3                 |                                         |                                           |                       |                    |                |                      |                    |
| 4                 |                                         |                                           |                       |                    |                |                      |                    |
| 5                 |                                         |                                           |                       |                    |                |                      |                    |
| Total Marks (100) |                                         |                                           |                       | Total Marks ( 50)  |                |                      |                    |

## Annexure A

### Industry Problem Solving Task

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

#### 1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions: • If

latency  $> 100$  ms  $\rightarrow$  system fails

- If accuracy  $< 85\%$   $\rightarrow$  unacceptable
- Hardware supports only moderate computation

Models available:

Viola-Jones: 78%, low latency

YOLO: 92%, moderate latency

Deep Learning: 95%, high latency

**Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.**

**ANSWER:**

**YOLO should be selected.** Viola-Jones fails because its accuracy is only 78%, which is below the required 85%. Deep Learning provides higher accuracy but requires high computation, which is unsuitable for the available hardware. YOLO provides **92% accuracy with moderate computation and latency**, making it the best choice for the surveillance system.

#### 2. Autonomous Vehicle Detection Logic

A vehicle detection system must:

- Prioritize safety (accuracy  $> 90\%$ )
- Maintain response time  $< 50$  ms

Options:

Method A: 88% accuracy, 30 ms

Method B: 93% accuracy, 70 ms

Method C: 91% accuracy, 45 ms

**Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.**

**ANSWER:**

**Method C should be selected.** Method A has a fast response time but its accuracy is only 88%, which does not satisfy the safety requirement. Method B provides 93% accuracy but takes 70 ms, exceeding the allowed response time. Method C provides **91% accuracy and 45 ms response time**, satisfying both conditions. Therefore, Method C is the safest and most appropriate choice.

#### 3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes

Motion Estimation: stable but less detailed

**Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.**

**ANSWER:**

**Motion Estimation should be selected for high-density traffic.** Optical Flow provides accurate motion information, but it can become noisy when many vehicles are present. Motion Estimation provides more stable results in complex and crowded scenes. Although it gives less detailed information, its stability makes it more suitable when robustness is the main requirement.

#### **4. Background Modeling Evaluation**

A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

System must maintain  $\geq 80\%$  accuracy across all scenarios

**Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.**

**ANSWER:**

**GMM alone does not meet the requirement.** GMM achieves 90% accuracy in static scenes, which satisfies the 80% requirement. However, its accuracy decreases to 65% in dynamic scenes, which is below the required level. Therefore, GMM should not be used alone. An **adaptive background modeling technique** can be added to improve performance in dynamic environments.

#### **5. Defect Detection System Decision**

Requirements:

- Detect subtle defects ( $>95\%$  accuracy)
- Adapt to product variations

Options:

Traditional methods: fast, less adaptive

Deep learning: accurate, requires training

**Compare both approaches and evaluate which meets requirements. Justify using performance vs adaptability logic.**

**ANSWER:**

**Deep Learning should be selected.** Traditional methods are fast but are less adaptive to changes in product characteristics and may not detect subtle defects effectively. Deep Learning can learn complex defect patterns from training data and provide high detection accuracy. Therefore, it is more suitable for achieving the required **more than 95% accuracy** while adapting to product variations.

**RUBRICS FOR CALCULATION:**

| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
|-------------------------------------|-----------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Understanding of Industry Problem   | 20%       | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25%       | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25%       | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15%       | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10%       | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |
| Professionalism                     | 5%        | Highly professional, well documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation               |